

VTU Belagavi

REGULAR EXAMINATION ? 2024

Renewable Energy Systems ? Regular Examination, 2024

Subject Code: EE601 Semester: 6 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Solar Energy Systems)

1. Explain Solar Energy Systems with a neat diagram/sketch and a suitable example. [12 Marks]
2. Discuss the engineering applications of Solar Energy Systems. [8 Marks]
3. Explain Solar Energy Systems with a neat diagram/sketch and a suitable example. [12 Marks]

SECTION B (Unit 2: Wind Energy Conversion)

4. Define Wind Energy Conversion. Explain its significance in Renewable Energy Systems with a suitable example. [8 Marks]
5. Derive the expression related to Wind Energy Conversion and explain each term. [8 Marks]
6. Solve a problem related to Wind Energy Conversion. Show all the steps clearly. [12 Marks]

SECTION C (Unit 3: Biomass and Fuel Cells)

7. Solve a problem related to Biomass and Fuel Cells. Show all the steps clearly. [8 Marks]
8. Describe the working principle of Biomass and Fuel Cells with necessary equations. [12 Marks]
9. How is Biomass and Fuel Cells implemented? Give an algorithm / procedure and explain. [12 Marks]

SECTION D (Unit 4: Energy Storage Systems)

10. Solve a problem related to Energy Storage Systems. Show all the steps clearly. [12 Marks]
11. Compare and contrast the key concepts involved in Energy Storage Systems. [12 Marks]
12. How is Energy Storage Systems implemented? Give an algorithm / procedure and explain. [12 Marks]

SECTION E (Unit 5: Hybrid Energy Systems)

13. Describe the working principle of Hybrid Energy Systems with necessary equations.

[12 Marks]

14. Discuss the engineering applications of Hybrid Energy Systems. [12 Marks]

15. List the advantages and limitations of Hybrid Energy Systems. Explain any two in detail. [12 Marks]

*** End of Question Paper ***